F79SP Session 7: Other Variants of Random Walks (2)

We study two further methods of dealing with random walks by comparing to SRW.

Coupling argument

Consider a random walk $\{S_n\}_{n\geq 0}$ with $a > 0$. Suppose that we know in comparison with a SRW (with or without zero jumps), says, $\{S'_n\}_{n\geq 0}$, we have:

- $S_n \geq S'_n$ for every n, then the ruin probability of $\{S_n\}_{n\geq 0}$ is bounded above by the ruin probability of $\{S'_n\}_{n\geq 0}$.
- $S_n \leq S'_n$ for every n, then the ruin probability of $\{S_n\}_{n\geq 0}$ is bounded below by the ruin probability of $\{S'_n\}_{n\geq 0}$.

Curiosity: For random variables A and B, $A \leq B$ (in the "usual stochastic order"), if $\mathbf{P}(A > x) \leq \mathbf{P}(B > x)$ for every x.

Using these arguments, we can construct upper/lower bounds for ruin probability of the random walk involved.

See Example in notes, pg3.12.

Scaling

Let $\{S_n\}_{n\geq 0}$ be a random walk and $c > 0$. Suppose that there exists a SRW on $\mathbb{Z}$ (with or without zero jumps) $\{S'_n\}_{n\geq 0}$, such that $S_n = cS'_n$ for every n. Then the ruin probability of $\{S_n\}_{n\geq 0}$ is the same as of $\{S'_n\}_{n\geq 0}$.

Question: Consider the random walk $\{S_n\}_{n\geq 0}$ given by $S_0 = 35$ and for $n \geq 1$, $S_n = 35 + \sum_{i=1}^{n} \xi_i$ where ξ_i's are i.i.d. random variable with:

$$\mathbf{P}(\xi_i = 7) = 0.8, \mathbf{P}(\xi_i = 0) = 0.1, \mathbf{P}(\xi_i = -7) = 0.1.$$

Find the ruin probability of $\{S_n\}_{n\geq 0}$.

Combination of coupling and scaling arguments

Example. Consider a random walk $\{S_n\}_{n\geq 0}$ with initial value $S_0 = 4$ and with i.i.d. increments having the following distribution:

$$\begin{aligned} \mathbf{P}(\xi_i = 2) &= p, \\ \mathbf{P}(\xi_i = 0) &= 1 - (p+q), \\ \mathbf{P}(\xi_i = -1) &= q, \end{aligned}$$

Provide two upper bounds for the ruin probability, using two different couplings.

Use of simulation

Sometimes probabilities of complex events, e.g. *ruin*, need to be estimated by simulation.

Suppose that, in a large number n of simulations of a given process, an event of interest occurs a total of k times. Then k has a binomial distribution with parameters n and p where p is the true (unknown) probability of the event.

We thus estimate p by

$$\hat{p} = k/n.$$

The *standard error* (i.e. *standard deviation*) of our estimate is

$$\begin{aligned} \mathrm{SE}(\hat{p}) &= \frac{1}{n}\mathrm{SD}(k) \\ &= \frac{1}{n}\sqrt{np(1-p)} \\ &\approx \frac{1}{n}\sqrt{n\hat{p}(1-\hat{p})} \\ &= \sqrt{\frac{k(n-k)}{n^3}}. \end{aligned}$$

Using the normal approximation to the binomial distribution, an approximate 95% confidence interval for p is given by

$$(\hat{p} - 2\,\mathrm{SE}(\hat{p}),\ \hat{p} + 2\,\mathrm{SE}(\hat{p})).$$